

CSC4082 – Group Project 05

Overview

This project is on implementing the k-means algorithm.

Steps

- Go through the research article “*Data Clustering: 50 Years Beyond K-Means*” by Prof. Anil K. Jain.
- Use k-means clustering to cluster the data on iris flowers given (iris.txt), to 3 clusters according to the following four numerical attributes: sepal length, sepal width, petal length, and petal width (First four columns of the file)
- Produce a report containing the following:
 - Implementation of K-means
 - Results
 - Show graphs of intermediate and final results and discuss
 - Answers to the following questions
 - How well does the resulting clustering match the species? (last column on the data file)
 - What alternatives can be used for reducing the sensitivity to initialization?



Image source – irisabramson.com

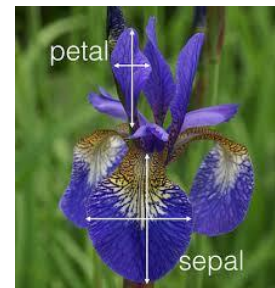


Image source – concrete interactive.com

Submission

Submit the Python file(s) with the original and resulting images together with a brief account of how the methods work.

Submission Deadline